Denusha Thavaruban

IT Undergraduate | Software Engineer Intern | Java, Spring Boot, React

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LinkedIn | GitHub | Portfolio

SUMMARY

Third-year IT & Management undergraduate at the University of Moratuwa (CGPA: 3.53/4.0) with hands-on experience developing full-stack applications, enterprise workflows, REST APIs, and database-driven systems using Java, Spring Boot, Python, TypeScript, React/Next.js, and PostgreSQL. Experienced in backend development, authentication, role-based access control, real-time systems, database design, and automated testing, with practical exposure to Docker, CI/CD, and cloud technologies. Demonstrated problem-solving, teamwork, and leadership through academic projects and serving as President of a Village Students' Union for four years. Seeking a Software Engineering internship to contribute to building scalable, secure, and maintainable software solutions.

EDUCATION

BSc (Hons) in Information Technology and Management **2024 - 2028**
SGPA : 3.53
Faculty of Information Technology, University of Moratuwa.

GCE Advanced Level **2020 - 2022**
Combined Mathematics | J / Vembadi Girls' High School

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, C
Backend & Frameworks: Spring Boot, NestJS, FastAPI, Node.js, REST APIs, Prisma ORM, Spring Data JPA, Hibernate
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS
Databases & Security: PostgreSQL, MySQL, Supabase, Spring Security, JWT, BCrypt, Row-Level Security (RLS), Database Migrations
Real-Time Technologies: Socket.IO, Server-Sent Events (SSE)
Cloud & DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, Render, Cloudinary
Testing & API Tools: Pytest, Playwright, Postman, API Testing, Automated Testing
Engineering Practices: Object-Oriented Programming (OOP), Modular Architecture, Role-Based Access Control (RBAC), Database Design, Git/GitHub, Agile Development

PROJECTS

QueueSense — Clinic Queue Management & Wait-Time Analytics Platform

GitHub

Full-Stack Development | Machine Learning

- Engineered a full-stack clinic queue management platform supporting patient check-ins, appointment scheduling, clinician workflows, and private queue tracking using synthetic healthcare operations data.
- Implemented role-based dashboards, QR-based self-check-in, real-time queue updates (SSE), operational analytics, and automated PDF/CSV reporting.
- Developed explainable Random Forest prediction models using SHAP, achieving 7.10-minute MAE against a 9.62-minute baseline on a synthetic test dataset.
- Implemented PostgreSQL-backed workflows, secure session authentication, audit logging, and automated database, API, and end-to-end testing.

Tech Stack: Next.js, React, TypeScript, Python, FastAPI, PostgreSQL, scikit-learn, SHAP, SSE, Docker Compose, GitHub Actions, Pytest, Playwright.

Performance Management System (PMS) — Employee Evaluation & Approval Workflow

GitHub

- Developed as part of a 5-member team, employee performance evaluation and multi-level approval workflows, including review, approval, rejection, and status tracking.
- Implemented My Team, Enquiry, and Notification features with search, filtering, performance scores, and feedback.
- Integrated Flask REST APIs with Supabase PostgreSQL and Azure AD SSO, supporting role-based organizational access.

Tech Stack: Next.js, React, TypeScript, Python, Flask, Supabase PostgreSQL, Azure AD SSO, REST APIs.

WayToHome — Full-Stack Bus Booking Platform GitHub

Full-Stack Development | Backend Engineering

- Engineered a bus booking platform featuring trip search, seat selection, reservation management, payment workflows, and QR-based e-ticket generation.
- Designed modular backend services using NestJS and Prisma for route management, booking operations, authentication, payments, and administration.
- Integrated Supabase authentication, JWT verification, RBAC, and server-side validation for secure passenger and administrator operations.
- Implemented real-time seat availability using Socket.IO and PostgreSQL transactional booking workflows to maintain reservation consistency.
- Developed administrative dashboards and containerized the application using Docker for deployment readiness.

Tech Stack: Next.js, NestJS, TypeScript, PostgreSQL, Prisma, Supabase, Socket.IO, Docker, CI/CD, AWS/Cloud Deployment.

Ink & Ideas — Full-Stack Blogging Platform GitHub

- Developed a responsive blogging platform with article scheduling, Markdown editing, search, bookmarks, comments, reactions, author following, and notifications.
- Built author analytics and admin dashboards for readership tracking, user management, and content moderation.
- Implemented secure authentication using Spring Security, JWT, and BCrypt, with Cloudinary media integration.
- Configured Docker Compose deployment with persistent PostgreSQL storage, API health checks, and environment-based secrets.

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Java 21, Spring Boot, Spring Security, JWT, PostgreSQL, Hibernate, Recharts, Cloudinary, Docker.

COURSES AND CERTIFICATIONS

Java - HackerRank | 2026 • Problem Solving - HackerRank | 2026 • Web Programming - University of Moratuwa | 2024 • Python Programming - University of Moratuwa | 2024 • SQL (Intermediate) - SoloLearn | 2024 • C (Intermediate) - SoloLearn | 2024

LEADERSHIP AND VOLUNTEERING

President — Village Students' Union **2019 – 2023**

- Led student activities and coordinated community initiatives.

Election Volunteer **2025**

- Assisted with election-related activities and volunteer coordination.

Community Volunteer — Grama Sevaka Office **2021**

- Supported community assistance and administrative activities during COVID-19.

Batch Coordinator — Tamil Literary Association **2024 – 2025**

- Coordinated batch activities and supported literary and cultural events.

REFERENCES

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